



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	CodeInsight AI: AI-Powered Code Review & Bug Detection	
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Abstract

CodeInsight AI is an AI-powered web application designed to assist software developers in reviewing source code and identifying potential bugs and code quality issues. Manual code review can be time-consuming and may overlook certain problems, especially when developers work with frequently changing code. The proposed system uses Artificial Intelligence to analyze source code and provide understandable feedback, helping developers identify issues and improve their code more efficiently.

The system allows users to upload source code or connect a GitHub repository for analysis. The submitted code is processed by the backend and analyzed using an AI model to identify potential bugs, coding issues, and areas for improvement. The system provides useful suggestions along with a code quality score and a summary of the review. Review results can also be stored so that users can access their previous analyses through a dashboard.

The application is developed using React.js for the frontend, FastAPI and Python for the backend, SQLite for data storage, Gemini AI API for intelligent code analysis, and GitHub API for repository integration. The project demonstrates how Artificial Intelligence can be applied to the software development process to reduce manual effort, improve code quality, and help developers perform faster and more consistent code reviews.

Signature of the Guide

HOD